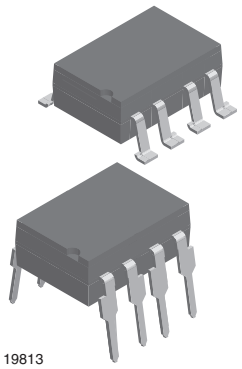
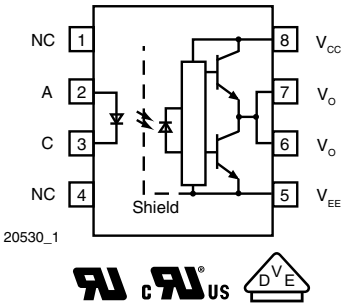
# 2.5 A Output Current IGBT and MOSFET Driver




## FEATURES

- 2.5 A minimum peak output current
- 25 kV/µs minimum common mode rejection (CMR) at $V_{CM}$ = 1500 V
- $I_{CC}$ = 2.5 mA maximum supply current
- Under voltage lock-out (UVLO) with hysteresis
- Wide operating $V_{CC}$ range: 15 V to 32 V
- 0.2 µs maximum pulse width distortion
- Industrial temperature range: - 40 °C to 110 °C
- 0.5 V maximum low level output voltage ($V_{OL}$)
- Reinforced insulation rated per DIN EN 60747-5-2
- Material categorization: For definitions of compliance please see www.vishay.com/doc?99912

Pb-free
e3
RoHS COMPLIANT

## DESCRIPTION

The VO3120 consists of a LED optically coupled to an integrated circuit with a power output stage. This optocoupler is ideally suited for driving power IGBTs and MOSFETs used in motor control inverter applications. The high operating voltage range of the output stage provides the drive voltages required by gate controlled devices. The voltage and current supplied by this optocoupler makes it ideally suited for directly driving IGBTs with ratings up to 800 V/50 A. For IGBTs with higher ratings, the VO3120 can be used to drive a discrete power stage which drives the IGBT gate.

## APPLICATIONS

- Isolated IGBT/MOSFET gate driver
- AC and brushless DC motor drives
- Induction stove top
- Industrial inverters
- Switch mode power supplies (SMPS)
- Uninterruptible power supplies (UPS)

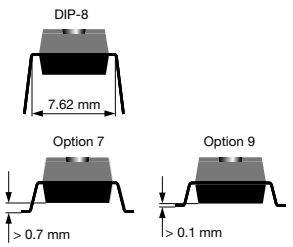
## AGENCY APPROVALS

- UL - file no. E52744 system code H, double protection
- cUL - file no. E52744, equivalent to CSA bulletin 5A
- DIN EN 60747-5-2 (VDE 0884) and reinforced insulation rating available with option 1

## ORDERING INFORMATION

| V | O | 3 | 1 | 2 | 0 | x | - | X | 0 | # | # | T |
|---|---|---|---|---|---|---|---|---|---|---|---|---|

PART NUMBER — PACKAGE OPTION — TAPE AND REEL




| PACKAGE | UL, cUL | UL, cUL, VDE |
|---|---|---|
| DIP-8, tubes | VO3120 | VO3120-X001 |
| SMD-8, option 7, tape and reel | VO3120-X007T | - |
| SMD-8, option 9, tape and reel | - | VO3120-X019T |


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| TRUTH TABLE | | | |
|---|---|---|---|
| LED | $V_{CC}$ - $V_{EE}$ "POSITIVE GOING" (TURN ON) | $V_{CC}$ - $V_{EE}$ "NEGATIVE GOING" (TURN OFF) | $V_O$ |
| Off | 0 V to 32 V | 0 V to 32 V | Low |
| On | 0 V to 11 V | 0 V to 9.5 V | Low |
| On | 11 V to 13.5 V | 9.5 V to 12 V | Transition |
| On | 13.5 V to 32 V | 12 V to 32 V | High |

| ABSOLUTE MAXIMUM RATINGS ($T_{amb}$ = 25 °C, unless otherwise specified) | | | | |
|---|---|---|---|---|
| PARAMETER | TEST CONDITION | SYMBOL | VALUE | UNIT |
| **INPUT** | | | | |
| Input forward current | | $I_F$ | 25 | mA |
| Peak transient input current | < 1 µs pulse width, 300 pps | $I_{F(TRAN)}$ | 1 | A |
| Reverse input voltage | | $V_R$ | 5 | V |
| Output power dissipation | | $P_{diss}$ | 45 | mW |
| **OUTPUT** | | | | |
| High peak output current (1) | | $I_{OH(PEAK)}$ | 2.5 | A |
| Low peak output current (1) | | $I_{OL(PEAK)}$ | 2.5 | A |
| Supply voltage | | ($V_{CC}$ - $V_{EE}$) | 0 to + 35 | V |
| Output voltage | | $V_{O(PEAK)}$ | 0 to + $V_{CC}$ | V |
| Output power dissipation | | $P_{diss}$ | 250 | mW |
| **OPTOCOUPLER** | | | | |
| Isolation test voltage (between emitter and detector) | t = 1 s | $V_{ISO}$ | 5300 | $V_{RMS}$ |
| Storage temperature range | | $T_S$ | - 55 to + 125 | °C |
| Ambient operating temperature range | | $T_A$ | - 40 to + 110 | °C |
| Total power dissipation | | $P_{tot}$ | 295 | mW |
| Lead solder temperature (2) | For 10 s, 1.6 mm below seating plane | | 260 | °C |

**Notes**

- Stresses in excess of the absolute maximum ratings can cause permanent damage to the device. Functional operation of the device is not implied at these or any other conditions in excess of those given in the operational sections of this document. Exposure to absolute maximum ratings for extended periods of the time can adversely affect reliability.

(1) Maximum pulse width = 10 µs, maximum duty cycle = 0.2 %. This value is intended to allow for component tolerances for designs with $I_O$ peak minimum = 2.5 A. See applications section for additional details on limiting $I_{OH}$ peak.

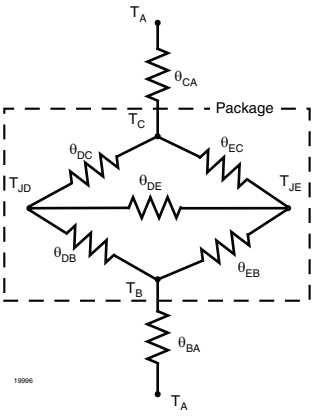
(2) Refer to reflow profile for soldering conditions for surface mounted devices (SMD). Refer to wave profile for soldering conditions for through hole devices (DIP).

| RECOMMENDED OPERATING CONDITION | | | | |
|---|---|---|---|---|
| PARAMETER | SYMBOL | MIN. | MAX. | UNIT |
| Power supply voltage | $V_{CC}$ - $V_{EE}$ | 15 | 32 | V |
| Input LED current (on) | $I_F$ | 7 | 16 | mA |
| Input voltage (off) | $V_{F(OFF)}$ | - 3 | 0.8 | V |
| Operating temperature | $T_{amb}$ | - 40 | + 110 | °C |


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## THERMAL CHARACTERISTICS

| PARAMETER | SYMBOL | VALUE | UNIT |
|---|---|---|---|
| LED power dissipation | $P_{diss}$ | 45 | mW |
| Output power dissipation | $P_{diss}$ | 250 | mW |
| Total power dissipation | $P_{tot}$ | 285 | mW |
| Maximum LED junction temperature | $T_{jmax.}$ | 125 | °C |
| Maximum output die junction temperature | $T_{jmax.}$ | 125 | °C |
| Thermal resistance, junction emitter to board | $\theta_{JEB}$ | 169 | °C/W |
| Thermal resistance, junction emitter to case | $\theta_{JEC}$ | 192 | °C/W |
| Thermal resistance, junction detector to board | $\theta_{JDB}$ | 82 | °C/W |
| Thermal resistance, junction detector to case | $\theta_{JDC}$ | 80 | °C/W |
| Thermal resistance, junction emitter to junction detector | $\theta_{JED}$ | 200 | °C/W |
| Thermal resistance, case to ambient | $\theta_{CA}$ | 2645 | °C/W |




**Note**

- The thermal characteristics table above were measured at 25 °C and the thermal model is represented in the thermal network below. Each resistance value given in this model can be used to calculate the temperatures at each node for a given operating condition. The thermal resistance from board to ambient will be dependent on the type of PCB, layout and thickness of copper traces. For a detailed explanation of the thermal model, please reference Vishay's Thermal Characteristics of Optocouplers application note.

## ELECTRICAL CHARACTERISTICS

| PARAMETER | TEST CONDITION | SYMBOL | MIN. | TYP. | MAX. | UNIT |
|---|---|---|---|---|---|---|
| High level output current | $V_O$ = ($V_{CC}$ - 4 V) | $I_{OH}$ (1) | 0.5 | | | A |
| | $V_O$ = ($V_{CC}$ - 15 V) | $I_{OH}$ (2) | 2.5 | | | A |
| Low level output current | $V_O$ = ($V_{EE}$ + 2.5 V) | $I_{OL}$ (1) | 0.5 | | | A |
| | $V_O$ = ($V_{EE}$ + 15 V) | $I_{OL}$ (2) | 2.5 | | | A |
| High level output voltage | $I_O$ = - 100 mA | $V_{OH}$ (3) | $V_{CC}$ - 4 | | | V |
| Low level output voltage | $I_O$ = 100 mA | $V_{OL}$ | | 0.2 | 0.5 | V |
| High level supply current | Output open, $I_F$ = 7 mA to 16 mA | $I_{CCH}$ | | | 2.5 | mA |
| Low level supply current | Output open, $V_F$ = - 3 V to + 0.8 V | $I_{CCL}$ | | | 2.5 | mA |
| Threshold input current low to high | $I_O$ = 0 mA, $V_O$ > 5 V | $I_{FLH}$ | | | 5 | mA |
| Threshold input voltage high to low | | $V_{FHL}$ | 0.8 | | | V |
| Input forward voltage | $I_F$ = 10 mA | $V_F$ | 1 | | 1.6 | V |
| Temperature coefficient of forward voltage | $I_F$ = 10 mA | $\Delta V_F/\Delta T_A$ | | - 1.4 | | mV/°C |
| Input reverse breakdown voltage | $I_R$ = 10 µA | $BV_R$ | 5 | | | V |
| Input capacitance | f = 1 MHz, $V_F$ = 0 V | $C_{IN}$ | | 60 | | pF |
| UVLO threshold | $V_O \geq 5$ V | $V_{UVLO+}$ | 11 | | 13.5 | V |
| | $I_F$ = 10 mA | $V_{UVLO-}$ | 9.5 | | 12 | V |
| UVLO hysteresis | | $UVLO_{HYS}$ | | 1.6 | | V |

**Notes**

- Minimum and maximum values were tested over recommended operating conditions ($T_A$ = - 40 °C to 110 °C, $I_{F(ON)}$ = 7 mA to 16 mA, $V_{F(OFF)}$ = - 3 V to 0.8 V, $V_{CC}$ = 15 V to 32 V, $V_{EE}$ = ground) unless otherwise specified. Typical values are characteristics of the device and are the result of engineering evaluations. Typical values are for information only and are not part of the testing requirements. All typical values were measured at $T_{amb}$ = 25 °C and with $V_{CC}$ - $V_{EE}$ = 32 V.

(1) Maximum pulse width = 50 µs, maximum duty cycle = 0.5 %.

(2) Maximum pulse width = 10 µs, maximum duty cycle = 0.2 %. This value is intended to allow for component tolerances for designs with $I_O$ peak minimum = 2.5 A.

(3) In this test $V_{OH}$ is measured with a dc load current. When driving capacitive loads $V_{OH}$ will approach $V_{CC}$ as $I_{OH}$ approaches zero A. Maximum pulse width = 1 ms, maximum duty cycle = 20 %.


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## TEST CIRCUITS

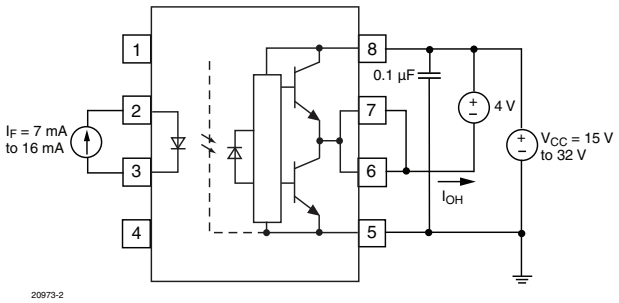



Fig. 1 - $I_{OH}$ Test Circuit

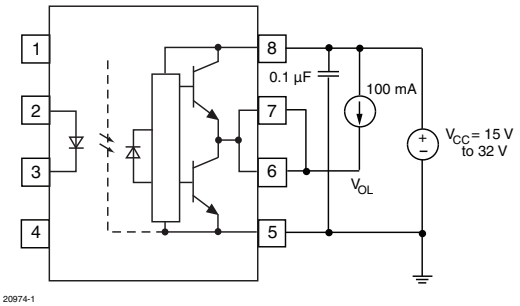



Fig. 4 - $V_{OL}$ Test Circuit

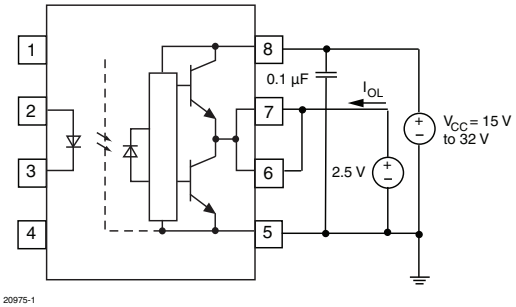



Fig. 2 - $I_{OL}$ Test Circuit

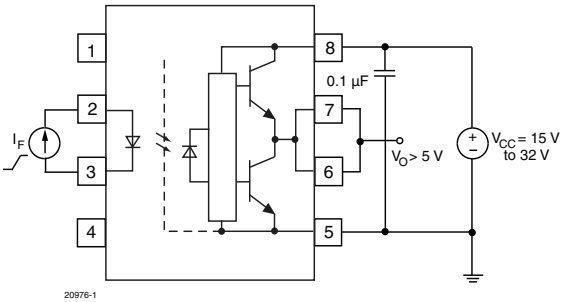



Fig. 5 - $I_{FLH}$ Test Circuit

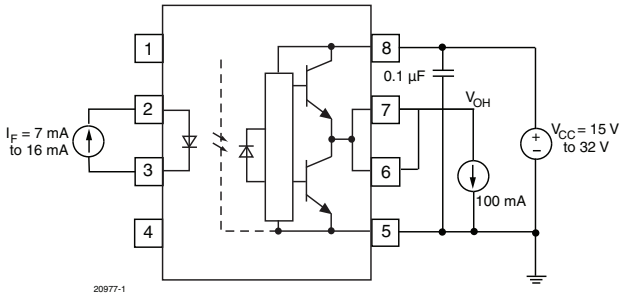



Fig. 3 - $V_{OH}$ Test Circuit

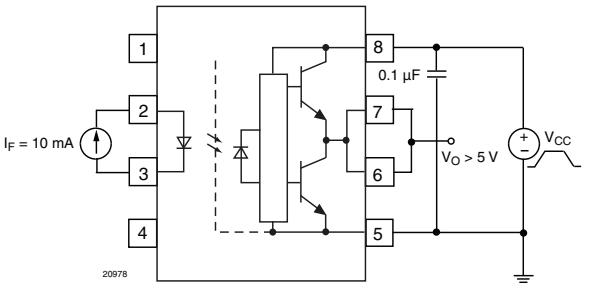



Fig. 6 - UVLO Test Circuit

## SWITCHING CHARACTERISTICS

| PARAMETER | TEST CONDITION | SYMBOL | MIN. | TYP. | MAX. | UNIT |
|---|---|---|---|---|---|---|
| Propagation delay time to logic low output (1) | $R_g$ = 10 Ω, $C_g$ = 10 nF, f = 10 kHz, duty cycle = 50 % | $t_{PHL}$ | 0.1 | | 0.4 | µs |
| Propagation delay time to logic high output (1) | $R_g$ = 10 Ω, $C_g$ = 10 nF, f = 10 kHz, duty cycle = 50 % | $t_{PLH}$ | 0.1 | | 0.4 | µs |
| Pulse width distortion (2) | $R_g$ = 10 Ω, $C_g$ = 10 nF, f = 10 kHz, duty cycle = 50 % | PWD | | | 0.2 | µs |
| Rise time | $R_g$ = 10 Ω, $C_g$ = 10 nF, f = 10 kHz, duty cycle = 50 % | $t_r$ | | 0.1 | | µs |
| Fall time | $R_g$ = 10 Ω, $C_g$ = 10 nF, f = 10 kHz, duty cycle = 50 % | $t_f$ | | 0.1 | | µs |
| UVLO turn on delay | $V_O$ > 5 V, $I_F$ = 10 mA | $T_{UVLO-ON}$ | | 0.8 | | µs |
| UVLO turn off delay | $V_O$ < 5 V, $I_F$ = 10 mA | $T_{UVLO-OFF}$ | | 0.6 | | µs |

**Notes**

(1) This load condition approximates the gate load of a 1200 V/75 A IGBT.

(2) Pulse width distortion (PWD) is defined as $|t_{PHL}-t_{PLH}|$ for any given device.

(3) The difference between $t_{PHL}$ and $t_{PLH}$ between any two VO3120 parts under the same test condition.



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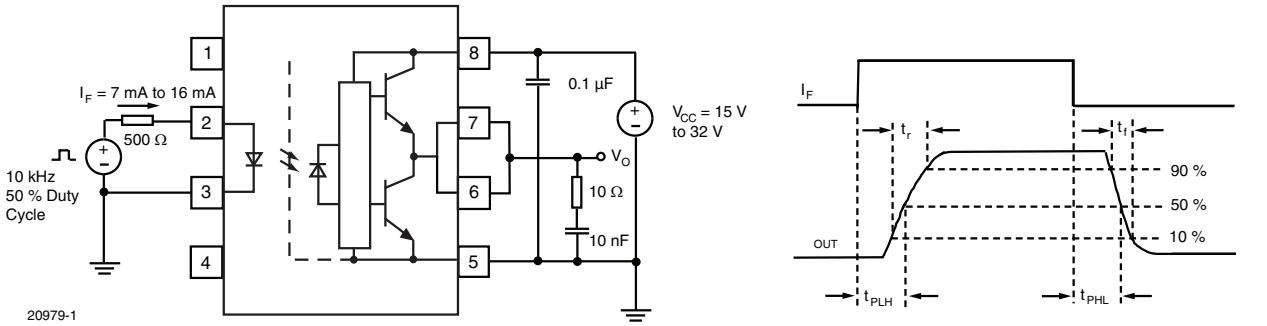



Fig. 7 - $t_{PLH}$, $t_{PHL}$, $t_r$ and $t_f$ Test Circuit and Waveforms

| COMMON MODE TRANSIENT IMMUNITY | | | | | | |
|---|---|---|---|---|---|---|
| **PARAMETER** | **TEST CONDITION** | **SYMBOL** | **MIN.** | **TYP.** | **MAX.** | **UNIT** |
| Common mode transient immunity at logic high output (1)(2) | $T_A$ = 25 °C, $I_F$ = 10 mA to 16 mA, $V_{CM}$ = 1500 V, $V_{CC}$ = 32 V | $\|CM_H\|$ | 25 | 35 | | kV/µs |
| Common mode transient immunity at logic low output (1)(3) | $T_A$ = 25 °C, $V_{CM}$ = 1500 V, $V_{CC}$ = 32 V, $V_F$ = 0 V | $\|CM_L\|$ | 25 | 35 | | kV/µs |

**Notes**

(1) Pins 1 and 4 need to be connected to LED common.

(2) Common mode transient immunity in the high state is the maximum tolerable $|dV_{CM}/dt|$ of the common mode pulse, $V_{CM}$, to assure that the output will remain in the high state (i.e., $V_O$ > 15 V).

(3) Common mode transient immunity in a low state is the maximum tolerable $|dV_{CM}/dt|$ of the common mode pulse, $V_{CM}$, to assure that the output will remain in a low state (i.e., $V_O$ < 1 V).

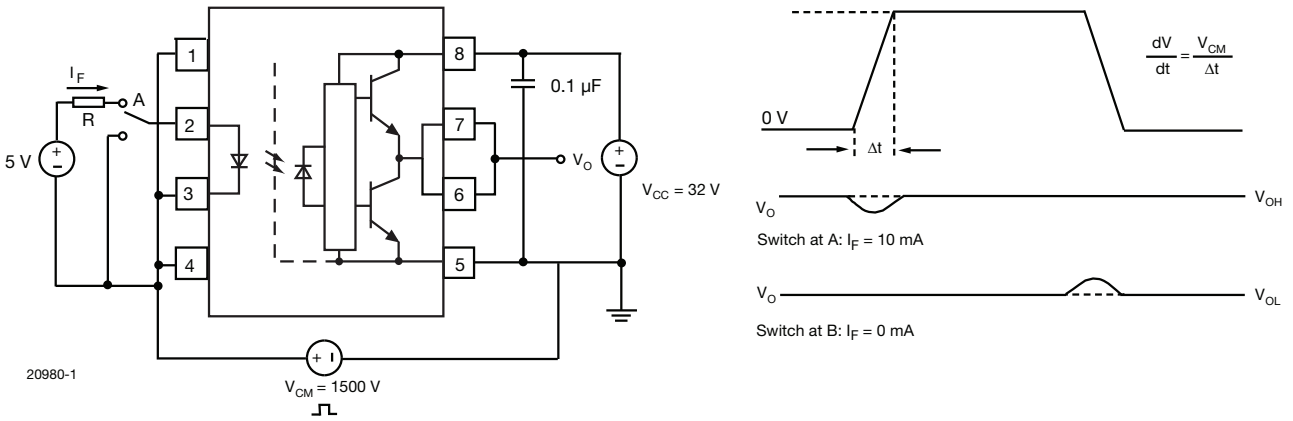



Fig. 8 - CMR Test Circuit and Waveforms


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| SAFETY AND INSULATION RATINGS | | | | | | |
|---|---|---|---|---|---|---|
| **PARAMETER** | **TEST CONDITION** | **SYMBOL** | **MIN.** | **TYP.** | **MAX.** | **UNIT** |
| Climatic classification (according to IEC 68 part 1) | | | | 40/110/21 | | |
| Comparative tracking index | | CTI | 175 | | 399 | |
| Peak transient overvoltage | | $V_{IOTM}$ | 8000 | | | V |
| Peak insulation voltage | | $V_{IORM}$ | 890 | | | V |
| Safety rating - power output | | $P_{SO}$ | | | 500 | mm |
| Safety rating - input current | | $I_{SI}$ | | | 300 | mm |
| Safety rating - temperature | | $T_{SI}$ | | | 175 | °C |
| Creepage distance | Standard DIP-8 | | 7 | | | mm |
| Clearance distance | Standard DIP-8 | | 7 | | | mm |
| Creepage distance | 400 mil DIP-8 | | 8 | | | mm |
| Clearance distance | 400 mil DIP-8 | | 8 | | | mm |

**Note**

- As per IEC 60747-5-2, §7.4.3.8.1, this optocoupler is reinforced rated and suitable for “safe electrical insulation” only within the safety ratings. Compliance with the safety ratings shall be ensured by means of protective circuits.

## TYPICAL CHARACTERISTICS ($T_{amb}$ = 25 °C, unless otherwise specified)

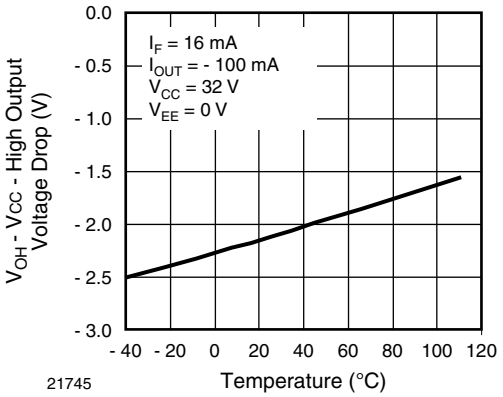



Fig. 9 - High Output Voltage Drop vs. Temperature

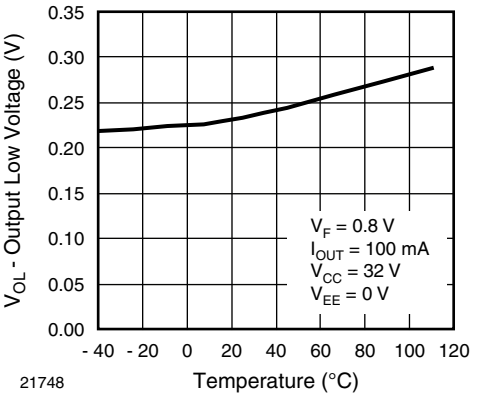



Fig. 11 - Output Low Voltage vs. Temperature

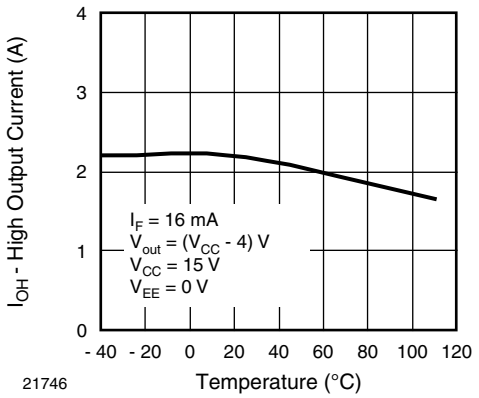



Fig. 10 - High Output Current vs. Temperature

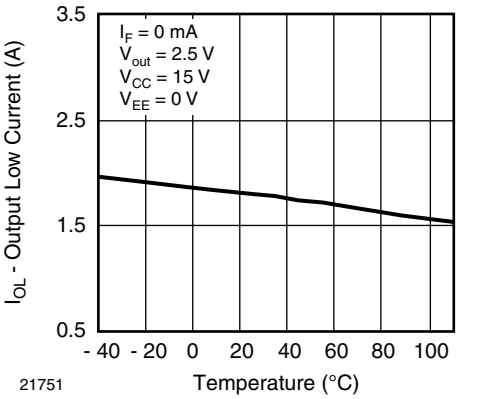



Fig. 12 - Output Low Current vs. Temperature


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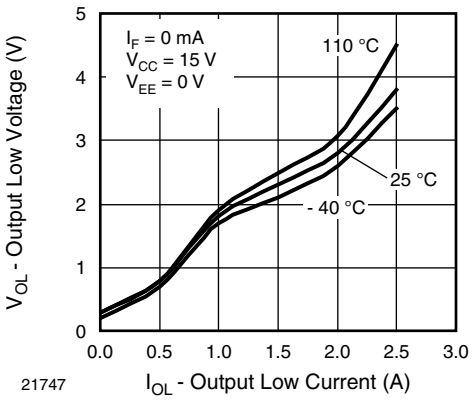



Fig. 13 - Output Low Voltage vs. Output Low Current

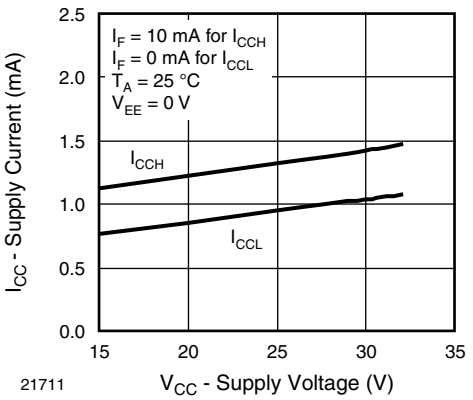



Fig. 16 - Supply Current vs. Supply Voltage

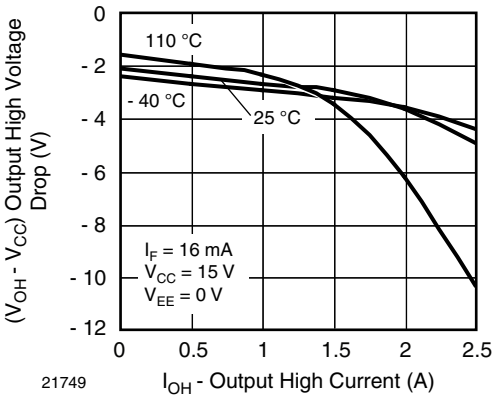



Fig. 14 - Output High Voltage Drop vs. Output High Current

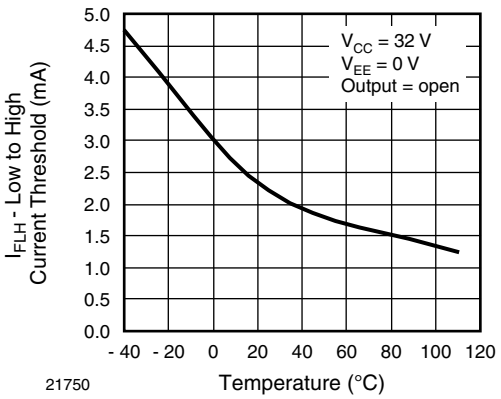



Fig. 17 - Low to High Current Threshold vs. Temperature

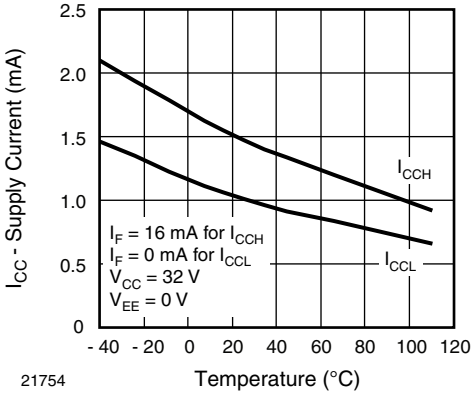



Fig. 15 - Supply Current vs. Temperature

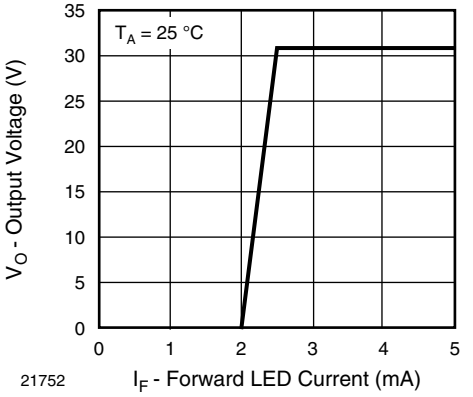



Fig. 18 - Transfer Characteristics


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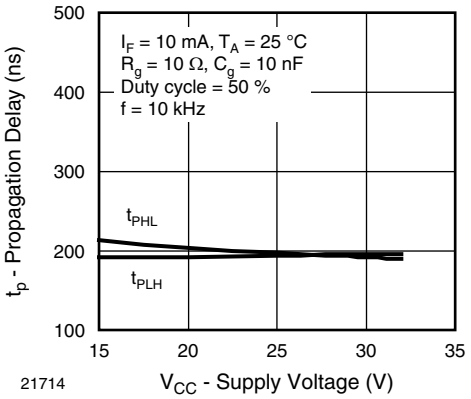



Fig. 19 - Propagation Delay vs. Supply Voltage

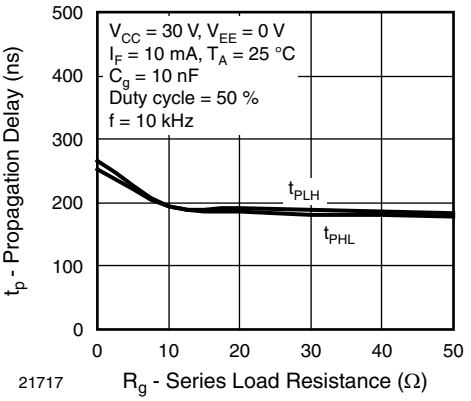



Fig. 22 - Propagation Delay vs. Series Load Resistance

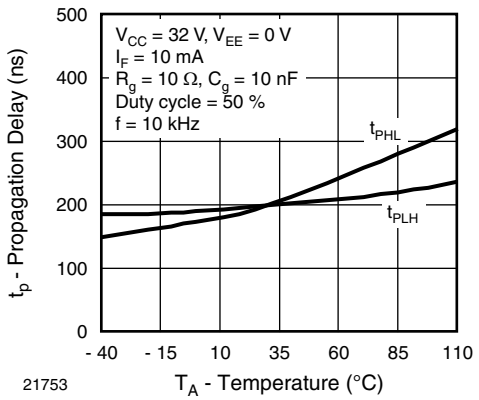



Fig. 20 - Propagation Delay vs. Temperature

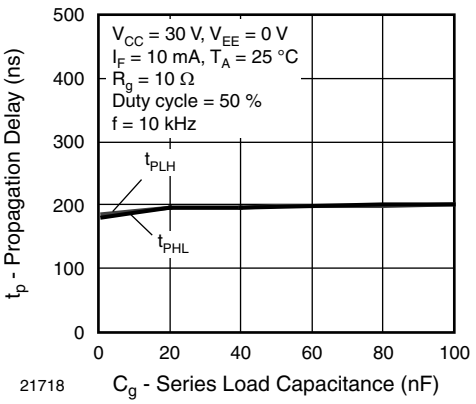



Fig. 23 - Propagation Delay vs. Series Load Capacitance

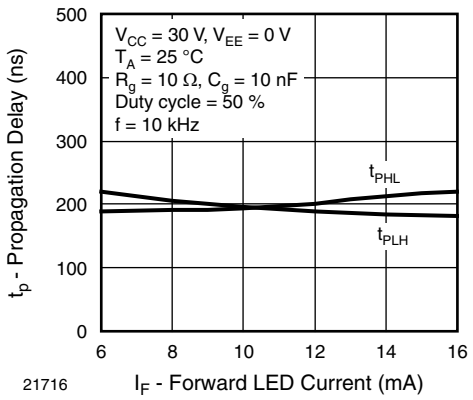
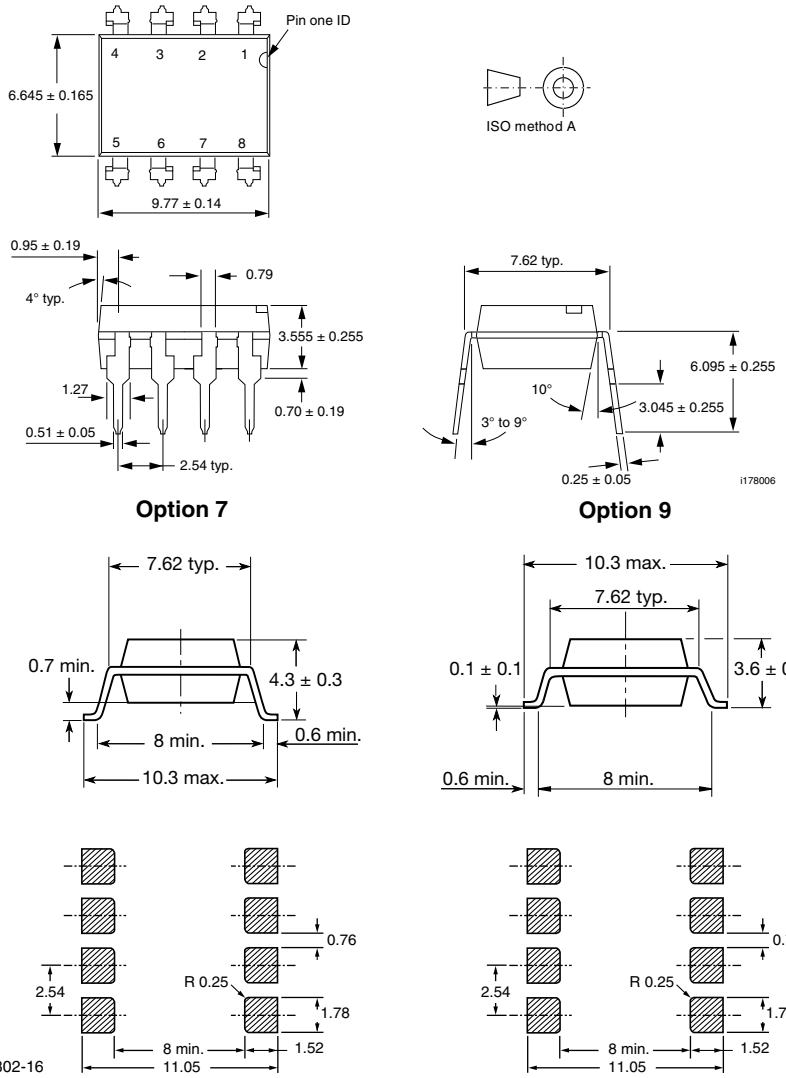


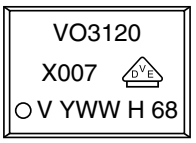

Fig. 21 - Propagation Delay vs. Forward LED Current


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## PACKAGE DIMENSIONS in millimeters




## PACKAGE MARKING




**Notes**

- The VDE logo is only marked on option 1 parts.
- Tape and reel suffix (T) is not part of the package marking.



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